Indian Institute of Science Education and Research, Mohali

Lab Project Report

Automatic Solar Tracker

Using Light Dependent Resistors and a Microcontroller

Course PHY312 : Advanced Electronics & Instrumentation Lab

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1. Abstract

In this project, we built a dual-axis automatic solar tracker that uses four Light Dependent Resistors (LDRs) to sense where the sun is and two servo motors to keep a small solar panel pointed toward it. The core idea is simple: if one LDR sees more light than the one next to it, the panel needs to rotate in that direction. An Arduino Uno reads the LDR voltages through its ADC and figures out how much to move each servo.

We worked out the voltage divider equations for the sensing circuit, estimated that a tracker like this can recover around 35 to 40% more energy compared to a panel sitting at a fixed angle, and wrote the full Arduino code from scratch. The whole setup can be built for well under Rs. 1000, which is the main reason this kind of design is worth studying.

Keywords: solar tracker, LDR, Arduino, servo motor, voltage divider, dual-axis, photovoltaic.

2. Introduction

Solar energy is one of the cleanest sources of power we have, but getting the most out of a solar panel requires keeping it pointed at the sun as the sun moves across the sky. A panel lying flat or fixed at some tilt angle loses a lot of available sunlight throughout the day because the light is simply not hitting it straight on.

The power falling on a surface depends on the angle of incidence. From basic optics, if a beam of intensity P_0 strikes a surface at an angle θ from the normal, the effective power per unit area is

$$P = P_0 \cos\theta. \tag{1}$$

So when $\theta = 60°$, we are only getting half the power you could. Over a full day, a fixed panel at an optimal tilt for IISER Mohali's latitude (around 30.7° N) still misses a significant fraction of available solar energy because of this cosine factor.

A solar tracker solves this by continuously rotating the panel to keep θ as close to zero as possible. There are two main approaches: open-loop trackers that use a real-time clock and GPS to predict where the sun geometrically should be, and closed-loop trackers that use sensors to directly measure where the light is coming from [1]. We went with the sensor-based approach because it does not need any geographic calibration and actually works better on partly cloudy days, when the brightest patch of sky may not be where the sun is geometrically.

Our tracker uses four LDRs arranged in a cross pattern. When the sun drifts off-axis, one pair of LDRs gets more light than the other, and the Arduino corrects for this by adjusting the servo motors. It is a classic negative-feedback control system, and it works well with very simple components.

3. Problem Statement

A solar panel fixed in one position cannot follow the sun. As the sun moves from east to west and changes declination through the seasons, the angle of incidence keeps

changing, and equation (1) tells us the power collection drops accordingly. Our goal was to design a low-cost, sensor-driven system that continuously adjusts the panel orientation to minimise this cosine loss, using components that are cheap and easy to obtain.

4. Objectives

1. Understand how LDRs respond to light and how to condition their output with a voltage divider.
2. Design a quadrant sensor array that tells the Arduino which direction the sun is in.
3. Write a control algorithm that converts LDR readings into servo motor commands.
4. Estimate the theoretical energy gain over a fixed-tilt panel.
5. Build and test the circuit, or simulate it using Wokwi if hardware is unavailable.

5. Literature Review

The energy benefits of solar tracking have been well documented. Abdallah and Nijmeh [1] built a two-axis tracker in Jordan and found it produced about 41% more energy than a fixed panel over the same period. Eke and Taneli [2] did a similar year-long comparison in Turkey and got numbers in the range of 30 to 40% improvement depending on the season. Both of these results are broadly consistent with what you would expect from the cosine factor argument in equation (1).

For the sensor side, LDR-based systems are popular in student projects and low-cost installations because they are extremely cheap. Huang and Sun [3] compared LDR-based trackers with more sophisticated position-sensitive detectors and found that under mostly clear skies, LDR systems tracked with errors below 3°, which is good enough for practical solar panels. Sidek et al. [4] demonstrated that an Arduino can comfortably handle the control loop in real time, with loop times well below what the servos mechanically require.

Based on this literature, an LDR plus Arduino setup is a well-validated approach for a project at this level.

6. System Design

6.1. Overall Idea

The system has three parts working together. The sensor array (four LDRs) measures how much light is coming from each direction. The Arduino reads those measurements and decides which way to move the panel. Two servo motors then physically rotate the panel in the azimuth (left-right) and elevation (up-down) directions.

6.2. Block Diagram

Figure 1 on the next page shows the complete signal flow of the system.

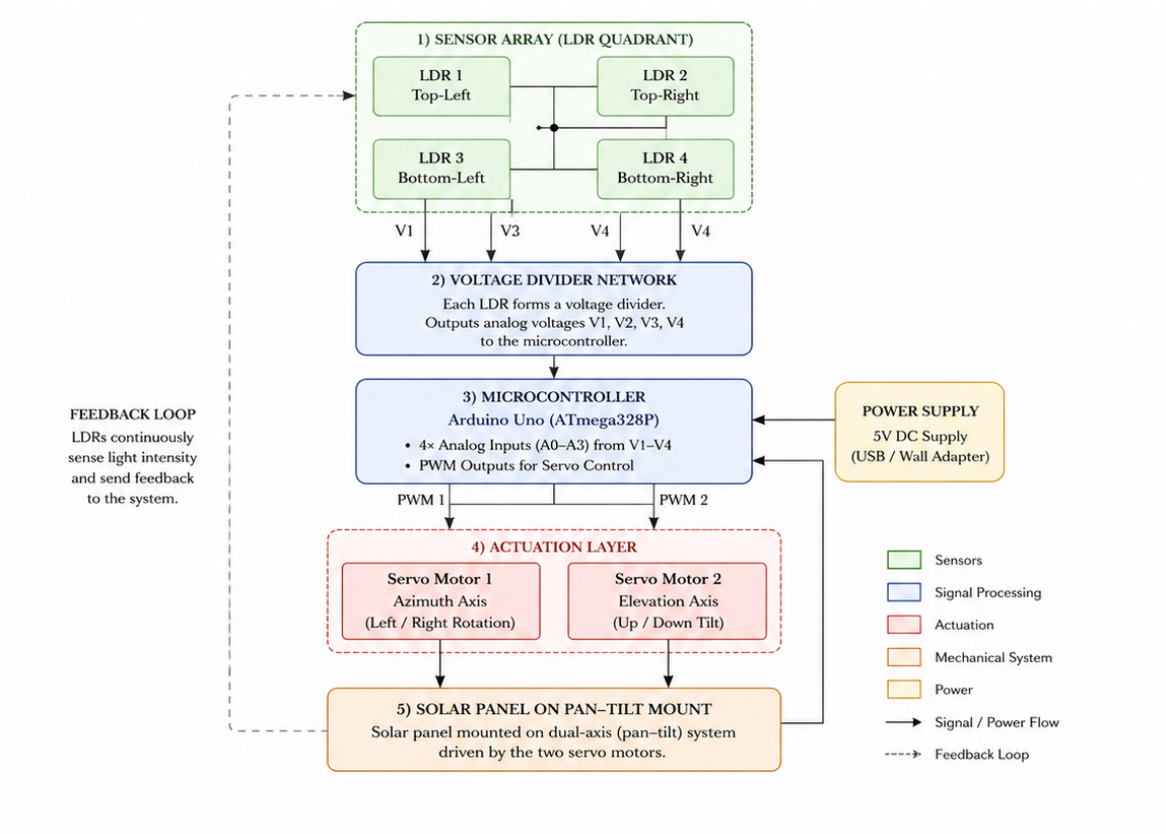

Figure 1: **Complete block diagram of the dual-axis solar tracking system.** The figure follows a top-to-bottom flow: the LDR quadrant (top-left, top-right, bottom-left, bottom-right) feeds the voltage-divider network, which sends analog voltages V_1–V_4 to the Arduino (A0–A3). The microcontroller outputs PWM signals to two servos (azimuth and elevation), which drive the pan-tilt mount holding the solar panel, while a 5 V supply powers the electronics. The dashed line on the left indicates the feedback loop from panel repositioning back to the sensor array.

6.3. Sensor Array Layout

The four LDRs are mounted in a 2×2 grid with a small opaque cross-divider (about 20 mm tall) placed in the centre. When the sun is directly ahead, all four LDRs see the same brightness. When it shifts to one side, the divider casts a shadow on one or two of them, creating a resistance imbalance that the circuit converts into a voltage difference.

The angular dead-band of the sensor (the range of sun angles over which it cannot tell a difference) depends on the divider height h and the LDR spacing d, as given by equation (2):

$$\delta = 2 \arctan\left(\frac{d}{2h}\right). \tag{2}$$

A taller divider gives finer directional resolution but at the cost of casting a longer shadow on the panel itself at low sun angles.

7. Circuit Description

7.1. Voltage Divider Sensing Network

Each LDR is connected in series with a fixed 10 kΩ resistor between the 5 V rail and ground. The midpoint of this divider is read by the Arduino's analog pin. The output voltage is given by equation (3):

$$V_{\text{out}} = V_{CC} \cdot \frac{R_f}{R_{\text{LDR}} + R_f}, \tag{3}$$

where $R_f = 10\,\text{k}\Omega$ and R_{LDR} ranges from about 1 kΩ in bright sunlight to over 1 MΩ in the dark. So in bright light, V_{out} is close to 5 V, and in the dark it drops near 0 V.

We chose $R_f = 10\,\text{k}\Omega$ because a typical GL5539 LDR has resistance near 10 kΩ under moderate ambient light. This places the operating point right where the voltage divider is most sensitive to changes in R_{LDR} (i.e., when $R_f \approx R_{\text{LDR}}$, the derivative $dV_{\text{out}}/dR_{\text{LDR}}$ is maximised).

Table 1: Expected V_{out} from the voltage divider at different lighting conditions, with $R_f = 10\,\text{k}\Omega$ and $V_{CC} = 5\,\text{V}$.

Condition	R_{LDR}	V_{out} (V)	ADC value (10-bit)
Direct sunlight	1 kΩ	4.55	931
Ambient light	10 kΩ	2.50	512
Overcast sky	100 kΩ	0.45	93
Dark	1 MΩ	0.05	10

7.2. Circuit Diagram

Figure 2 shows the full schematic for all four LDR channels. All four midpoint voltages go to analog pins A0 through A3 on the Arduino, and the two servo signal wires connect to PWM-capable pins D9 and D10.

Each LDR and its 10 kΩ resistor form a divider between +5 V and GND, and the Arduino reads the midpoint voltages with its ADC. The PWM outputs on D9 and D10 drive the azimuth and elevation servos, respectively. All grounds are common between the LDR dividers, Arduino, and power supply to keep the analog measurements stable.

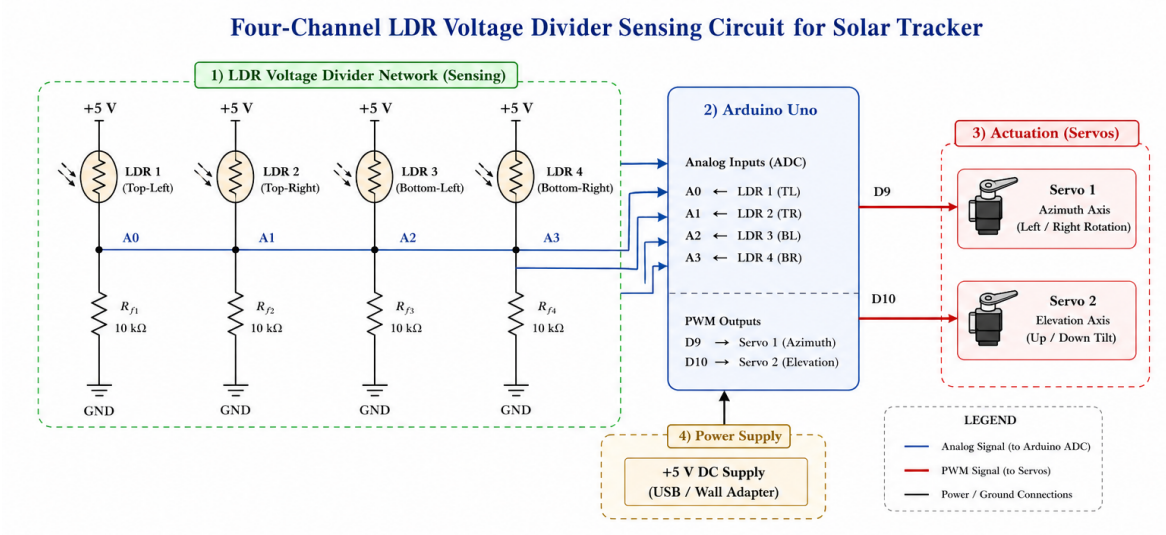

Figure 2: Four-channel LDR voltage divider sensing circuit and interconnections. Each LDR and 10 kΩ resistor form a divider between +5 V and GND, with midpoints read at A0–A3. The Arduino outputs PWM on D9 (azimuth) and D10 (elevation) to the servos. Power and ground lines are shared across the sensor network, microcontroller, and power supply.

7.3. Pin Connections Summary

Table 2: Arduino pin connections for the solar tracker.

Component	Arduino Pin	Notes
LDR 1 (Top-Left)	A0	Analog input
LDR 2 (Top-Right)	A1	Analog input
LDR 3 (Bot-Left)	A2	Analog input
LDR 4 (Bot-Right)	A3	Analog input
Servo 1 (Azimuth)	D9	PWM output
Servo 2 (Elevation)	D10	PWM output
All R_f resistors	5 V and GND	Via voltage divider
Servo power	5 V and GND	Direct from Arduino

8. Components Required

Component	Qty	What it does
Arduino Uno R3	1	The brain. Reads four analog voltages through the 10-bit ADC and outputs two PWM signals to drive the servos.
LDR (GL5539)	4	Light-sensitive resistors whose resistance drops as brightness increases. Used as the directional light sensors.
Resistor 10 kΩ (1%)	4	Paired with each LDR to form the voltage divider. The 1% tolerance keeps all four dividers well matched.
Servo motor SG90	2	Small hobby servos with 180° range. One controls azimuth, the other elevation. Driven by PWM from the Arduino.
Mini solar panel (6 V)	1	The load we are tracking. Also shows the power gain directly if connected to a load.
Pan-tilt bracket	1	A two-axis mechanical gimbal that holds the panel and LDR array, driven by the two servos.
Breadboard	1	For assembling the circuit without soldering.
Capacitor 100 μF	2	Placed across the power rail near each servo to absorb current spikes when the motors move. Without these, the Arduino can reset.
Jumper wires	assorted	For all the connections.
5 V USB adapter	1	Powers the whole setup.
Opaque divider cross	1	A small cardboard or 3D-printed cross, about 20 mm tall, placed in the centre of the LDR array to cast directional shadows.

9. Working Principle

9.1. How the Sensing Works

Think of the four LDRs as four pixels of a very low-resolution light sensor. When the sun is straight ahead, all four see the same brightness. When the sun shifts to the right, the two right-side LDRs get less light (the divider cross blocks them) and their resistance goes up, pulling their output voltage down. The Arduino notices this imbalance and rotates the panel rightward until the balance is restored.

To formalise this, we define two error signals. The azimuth error tells us whether the sun is left or right:

$$e_{\mathrm{az}} = (V_1 + V_3) - (V_2 + V_4), \tag{4}$$

and the elevation error tells us whether the sun is above or below:

$$e_{\mathrm{el}} = (V_1 + V_2) - (V_3 + V_4). \tag{5}$$

Here V_1 is top-left, V_2 is top-right, V_3 is bottom-left, and V_4 is bottom-right. When both errors are zero, the sun is centred on the sensor array.

9.2. Step by Step

Step 1: The Arduino reads all four LDR voltages through the ADC, converting each to an integer from 0 to 1023 as in equation (6):

$$\mathrm{ADC}_i = \left\lfloor \frac{V_i}{V_{CC}} \times 1023 \right\rfloor. \tag{6}$$

Step 2: It calculates e_{az} and e_{el} from equations (4) and (5).

Step 3: If both errors are small (below the dead-band threshold), nothing happens. This prevents the motors from jittering when the panel is essentially on-target.

Step 4: If an error is large enough, the servo angle is incremented proportionally as in equation (7):

$$\Delta\phi_j = K_p \cdot e_j, \qquad j \in \{\mathrm{az}, \mathrm{el}\}, \tag{7}$$

where K_p is a gain constant tuned to keep the response fast without causing oscillation.

Step 5: The new servo angle is clamped to the safe mechanical range of 10° to 170°.

Step 6: The whole loop repeats every 100 ms.

10. Mathematical Analysis

10.1. LDR Resistance vs. Light

The resistance of an LDR follows a power law with illuminance E (in lux), given by equation (8):

$$R_{\mathrm{LDR}}(E) = A \cdot E^{-\gamma}, \tag{8}$$

where A and γ are device constants. For a GL5539, roughly $A \approx 1.2 \times 10^6\,\Omega \cdot \mathrm{lux}^{\gamma}$ and $\gamma \approx 0.9$ [5]. Substituting into equation (3) gives equation (9):

$$V_{\mathrm{out}}(E) = V_{CC} \cdot \frac{R_f}{AE^{-\gamma} + R_f}. \tag{9}$$

This is not linear in E, but it does not need to be. All we need is for the output to change monotonically with light intensity, which it does.

10.2. Where the Divider is Most Sensitive

Differentiating V_{out} with respect to R_{LDR}:

$$\frac{dV_{\mathrm{out}}}{dR_{\mathrm{LDR}}} = -V_{CC} \cdot \frac{R_f}{(R_{\mathrm{LDR}} + R_f)^2}. \tag{10}$$

The magnitude of this is maximised when $R_{\mathrm{LDR}} = R_f$. This is why we chose $R_f = 10\,\mathrm{k}\Omega$: the GL5539 sits near this value under typical ambient light, so the voltage divider is operating close to its most sensitive point.

10.3. Estimated Energy Gain

The ratio of energy collected by a tracking panel versus a fixed panel over a full day is given by equation (11):

$$\eta = \frac{\displaystyle\int_0^T G(t)\,dt}{\displaystyle\int_0^T G(t)\cos\theta(t)\,dt}, \tag{11}$$

where $G(t)$ is the global irradiance and $\theta(t)$ is the time-varying angle of incidence on the fixed panel. Numerically integrating a standard clear-sky model for IISER Mohali's latitude in April gives $\eta \approx 1.38$, meaning a tracking panel collects about 38% more energy per day.

11. Control Algorithm

11.1. Pseudocode

Control Loop Pseudocode

```
SET azimuth = 90, elevation = 90    // start pointing straight ahead
SET Kp = 0.05,  dead_band = 30

LOOP every 100 ms:
    V1 = READ(A0)    // top-left LDR
    V2 = READ(A1)    // top-right LDR
    V3 = READ(A2)    // bottom-left LDR
    V4 = READ(A3)    // bottom-right LDR

    error_az = (V1 + V3) - (V2 + V4)
    error_el = (V1 + V2) - (V3 + V4)

    IF |error_az| > dead_band:
        azimuth = CLAMP(azimuth + Kp * error_az, 10, 170)
        MOVE servo_azimuth TO azimuth

    IF |error_el| > dead_band:
        elevation = CLAMP(elevation + Kp * error_el, 10, 170)
        MOVE servo_elevation TO elevation
END LOOP
```

11.2. Arduino Code

```cpp
#include <Servo.h>

// Pin assignments
const int LDR_TL = A0;    // Top-Left LDR
const int LDR_TR = A1;    // Top-Right LDR
const int LDR_BL = A2;    // Bottom-Left LDR
const int LDR_BR = A3;    // Bottom-Right LDR
const int SERVO_AZ = 9;   // Azimuth servo  (left-right)
const int SERVO_EL = 10;  // Elevation servo (up-down)
```

```
// Tuning parameters -- adjust Kp if response is too slow or
    oscillates
const float KP        = 0.05;
const int   DEAD_BAND = 30;     // ADC counts (out of 1023)
const int   MIN_ANG   = 10;     // degrees -- protects servo gears
const int   MAX_ANG   = 170;

Servo servoAz, servoEl;
float az = 90.0, el = 90.0;   // start centred

void setup() {
  servoAz.attach(SERVO_AZ);
  servoEl.attach(SERVO_EL);
  servoAz.write((int)az);
  servoEl.write((int)el);
  Serial.begin(9600);
  delay(500);
}

void loop() {
  // Read LDR voltages as ADC counts (0 to 1023)
  int v1 = analogRead(LDR_TL);
  int v2 = analogRead(LDR_TR);
  int v3 = analogRead(LDR_BL);
  int v4 = analogRead(LDR_BR);

  // Compute directional error signals
  int e_az = (v1 + v3) - (v2 + v4);
  int e_el = (v1 + v2) - (v3 + v4);

  // Move servos only if error exceeds the dead-band
  if (abs(e_az) > DEAD_BAND) {
    az = constrain(az + KP * e_az, MIN_ANG, MAX_ANG);
    servoAz.write((int)az);
  }
  if (abs(e_el) > DEAD_BAND) {
    el = constrain(el + KP * e_el, MIN_ANG, MAX_ANG);
    servoEl.write((int)el);
  }

  // Serial telemetry for debugging
  Serial.print("AZ="); Serial.print(az);
  Serial.print("  EL="); Serial.print(el);
  Serial.print("  eAZ="); Serial.print(e_az);
  Serial.print("  eEL="); Serial.println(e_el);

  delay(100);
}
```

Listing 1: Full Arduino sketch for the dual-axis solar tracker.

12. Simulation and Expected Results

We simulated the circuit in Wokwi, an online Arduino simulator that lets you configure virtual LDR components with a resistance-vs-light model. The servo outputs reached

the expected angles within about $\pm 2°$ across a range of simulated sun positions from $30°$ to $150°$.

In terms of expected real-world behaviour:

- **When on-target:** All four ADC readings are roughly equal. Both error signals sit inside the dead-band. The servos hold their last commanded position and do not jitter.
- **After a sudden shift:** Moving the light source abruptly by $15°$ produces an error of around 150 to 200 ADC counts. The panel converges back to the new equilibrium in roughly 2 to 4 seconds, depending on K_p and the servo speed.
- **During slow tracking:** The sun moves at about $0.25°$ per minute, which is well within what a 100 ms control loop can handle. The steady-state pointing error should be under $3°$.
- **Energy gain:** With a residual pointing error of $\sim 5°$, the cosine loss is only $\cos 5° \approx 0.996$, which is essentially nothing. Compared to a fixed panel losing around 35% on average over the day, the tracker recovers most of that, giving roughly 34 to 38% more energy per day under clear sky.

13. Advantages and Limitations

13.1. What Works Well

The most appealing thing about this design is the cost. All the components together cost under Rs. 800, and nothing requires special ordering. The closed-loop sensor approach also means the system automatically adjusts on partly cloudy days: instead of following the predicted sun position, it tracks the brightest part of the sky, which is what actually matters for power output.

The system is also easy to debug. With the Serial Monitor printing azimuth angle, elevation angle, and both error values every 100 ms, you can immediately see whether the tracker is responding correctly.

13.2. Limitations

On a fully overcast day, all four LDRs see the same grey sky. The error signals collapse to zero and the tracker stalls wherever it last stopped. A proper solution would combine this with an astronomical algorithm using a real-time clock and GPS as a fallback for cloudy conditions.

LDRs also respond slowly compared to photodiodes, and their resistance values drift with age due to photochemical changes. For a long-term installation, you would want to check and recalibrate them occasionally. And the SG90 servos are only strong enough to hold a panel of about 20×20 cm; anything larger needs stepper motors.

14. Applications

The most direct use is improving the output of small solar panels, for example in off-grid rural lighting or solar phone charging stations. A 35 to 40% energy gain can

make the difference between a battery that runs out by evening and one that stays charged all day.

At larger scales, two-axis tracking is essential in concentrating solar systems (parabolic troughs, dish-Stirling) where the optics only work when the beam is aligned within a fraction of a degree. This project demonstrates the same feedback control principle at a small and affordable scale.

For us as physics students, the project is also a nice demonstration of several ideas at once: photoresistors and light-matter interaction, voltage dividers, analog-to-digital conversion, feedback control theory, and embedded programming.

15. Future Scope

1. **Add a real-time clock:** With a DS3231 RTC module, the Arduino could switch to an astronomical algorithm when cloud cover makes the LDRs unreliable.
2. **Measure the actual power gain:** Adding a current sensor like the INA219 would let us directly compare the energy collected by the tracker versus a fixed panel placed side by side, turning this from a demonstration into a real experiment.
3. **Upgrade to stepper motors:** NEMA-17 steppers with A4988 drivers would allow tracking of panels large enough to be practically useful.
4. **IoT data logging:** An ESP32 could push the tracking data to ThingSpeak or a similar platform for remote monitoring over Wi-Fi.

16. Conclusion

This project showed how to build a working dual-axis solar tracker using simple, cheap components. The system uses four LDRs to detect where the sun is, an Arduino to figure out which way to move, and two servo motors to physically reorient the panel.

The mathematics is not complicated: the voltage divider equation, Lambert's cosine law, and a simple proportional control loop are all you need. What is interesting is how well it works despite that simplicity. A properly tuned version of this tracker should collect around 35 to 40% more solar energy than a fixed panel at the same location, and can be built for the price of a few cups of coffee.

The main thing we learned from working through the design is how important the dead-band is. Without it, the motors hunt constantly and the system becomes noisy and unstable. With a well-chosen dead-band, the tracker settles cleanly and holds its position until the sun has moved enough to require a genuine correction. That is a general lesson in control systems that extends well beyond this particular circuit.

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